

Comparitive Performance Report for tentacle-3-way-replica-balance-reads-off vs tentacle-3-way-replica-balance-reads-on vs direct-reads-3-way-replica-balance-reads- off vs direct-reads-3-way-replica-balance-reads- on

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Comparison summary for tentacle-3-way-replica-balance-reads-off vs tentacle-3-way-replica-balance-reads-on vs direct-reads-3-way-replica-balance-reads- off vs direct-reads-3-way-replica-balance-reads- on

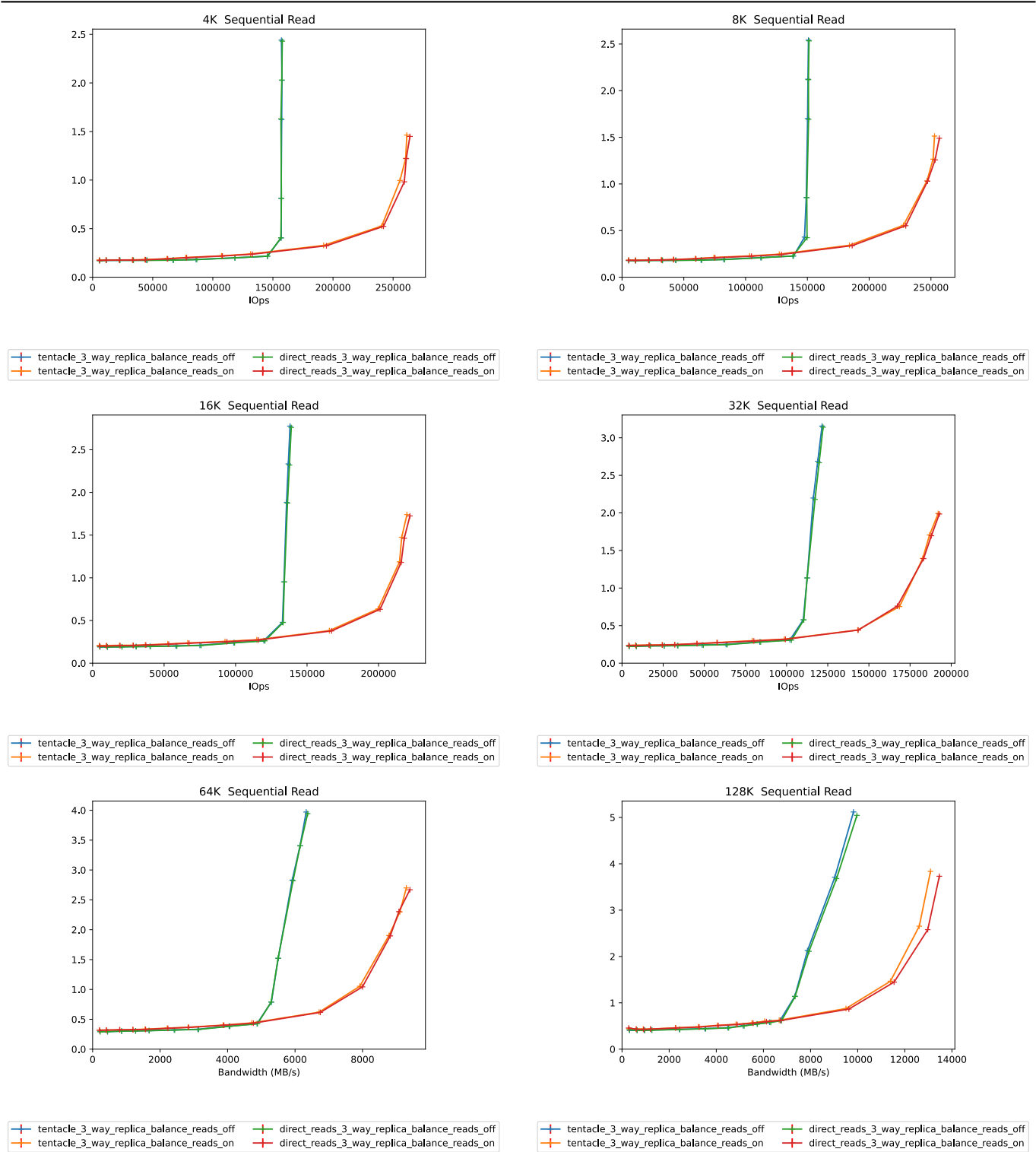
Sequential Read	tentacle_3_way-replica-balance-reads-off	tentacle_3_way-replica-balance-reads-on	%change	direct-reads-3-way-replica-balance-reads-off	direct-reads-3-way-replica-balance-reads-on	%change
4K	157548@2.0ms	261426@1.5	66%	157805@2.4	263998@1.4	68%
8K	150852@2.5ms	252900@1.5	68%	151340@2.5	256792@1.5	70%
16K	138108@2.8ms	219887@1.7	59%	138955@2.8	221977@1.7	61%
32K	121533@3.2ms	192102@2.0	58%	122155@3.1	192803@2.0	59%
64K	6331@4.0ms	9299@2.7	47%	6374@3.9	9405@2.7	49%
128K	9823@5.1ms	13091@3.8	33%	9968@5.0	13465@3.7	37%
256K	14295@7.0ms	17215@5.8	20%	14279@7.0	17337@5.8	21%
384K	14995@6.7ms	16934@11.9	13%	16025@12.6	16088@3.9	7%
512K	17061@15.7ms	14250@18.8	-16%	17311@15.5	14841@9.0	-13%
768K	17212@23.4ms	16851@11.9	-2%	17046@23.6	15478@8.1	-10%
1024K	16689@24.1ms	15398@17.4	-8%	16582@16.2	16202@16.6	-3%
2048K	14689@9.1ms	14321@28.1	-3%	14981@17.9	13606@9.8	-7%
4096K	13597@59.2ms	14110@57.1	4%	14188@37.8	12237@9.6	-10%

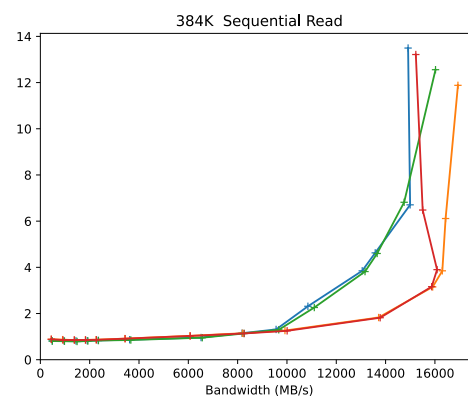
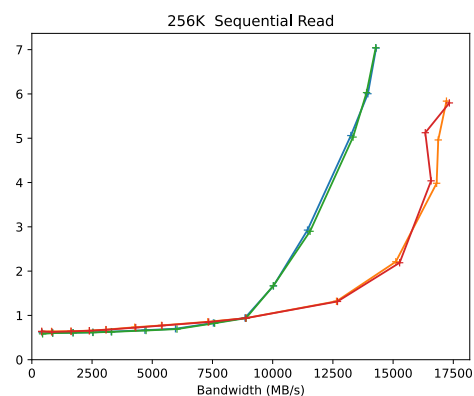
Random Read	tentacle_3_way-replica-balance-reads-off	tentacle_3_way-replica-balance-reads-on	%change	direct-reads-3-way-replica-balance-reads-off	direct-reads-3-way-replica-balance-reads-on	%change
4K	219491@1.2ms	239094@1.1	9%	223682@1.1	242195@1.3	10%
8K	209044@1.2ms	225342@1.7	8%	216775@1.8	235156@1.4	12%
16K	188007@1.4ms	203290@1.9	8%	193155@1.7	210385@1.5	12%
32K	167762@2.3ms	182074@1.8	9%	167800@1.9	185568@1.7	11%
64K	9032@2.8ms	9412@2.7	4%	8939@2.3	9559@2.2	6%
128K	13244@3.8ms	13212@2.5	-0%	13489@2.5	13258@3.8	0%
256K	16481@4.1ms	14106@7.1	-14%	16720@6.0	16375@6.1	-1%
384K	17128@11.7ms	16696@3.8	-3%	17456@5.8	16690@12.1	-3%
512K	17600@15.2ms	15960@4.2	-9%	17682@7.6	15909@8.4	-10%
768K	17110@23.5ms	16308@6.2	-5%	17132@23.5	16925@11.9	-1%
1024K	16507@16.2ms	16095@8.3	-2%	16383@16.4	14819@4.5	-10%
2048K	14724@18.2ms	14222@18.8	-3%	14846@18.1	13804@9.7	-6%
4096K	13969@57.6ms	13727@58.7	-2%	13878@58.0	12502@10.7	-11%

Random							
Read/Write	tentacle_3_way-replica-balance-reads-off	tentacle_3_way-replica-balance-reads-on	% change	direct-reads-3-way-replica-balance-reads-off	direct-reads-3-way-replica-balance-reads-on	% change	
4K_70/30	119218@3.2ms	123422@3.1	4%	122950@3.1	3%	128674@3.0	8%
16K_70/30	101882@3.8ms	106175@3.6	4%	103921@3.7	2%	110524@3.5	8%
64K_30/70	3170@5.3ms	2947@2.8	-7%	3282@7.6	4%	3125@5.3	-1%
64K_70/30	4847@4.3ms	4942@5.1	2%	4961@3.4	2%	5107@4.9	5%

Response Curves

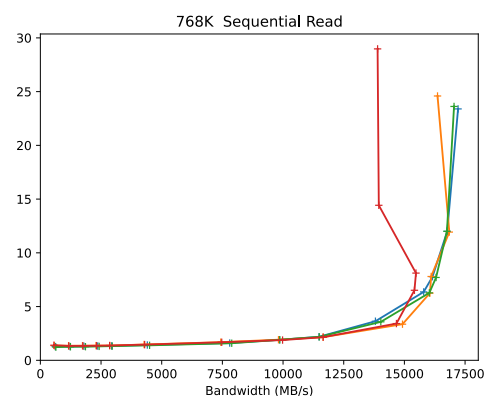
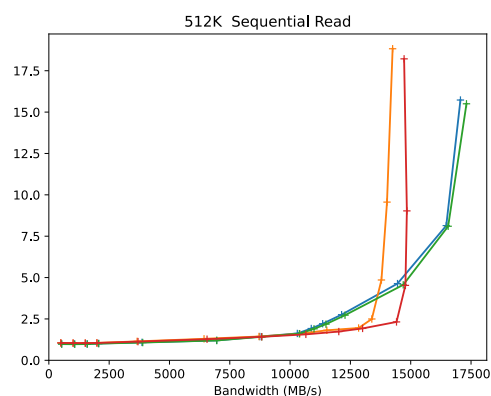
Sequential Read





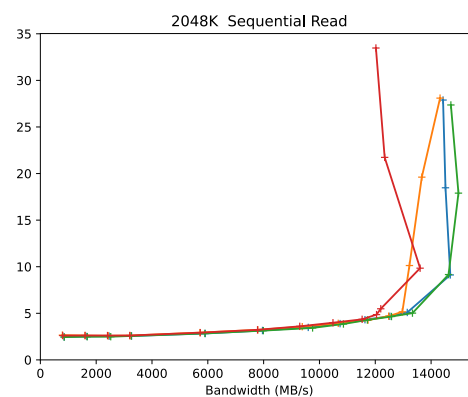
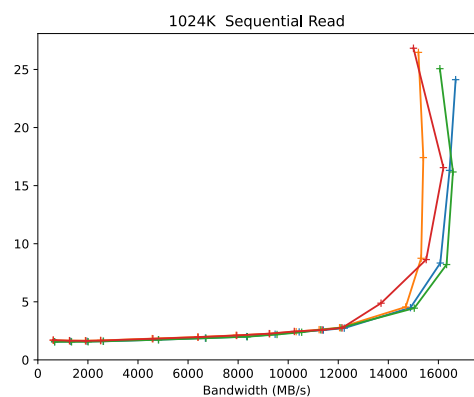
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off
tentacle_3_way_replica_balance_reads_on direct_reads_3_way_replica_balance_reads_on

tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off
tentacle_3_way_replica_balance_reads_on direct_reads_3_way_replica_balance_reads_on



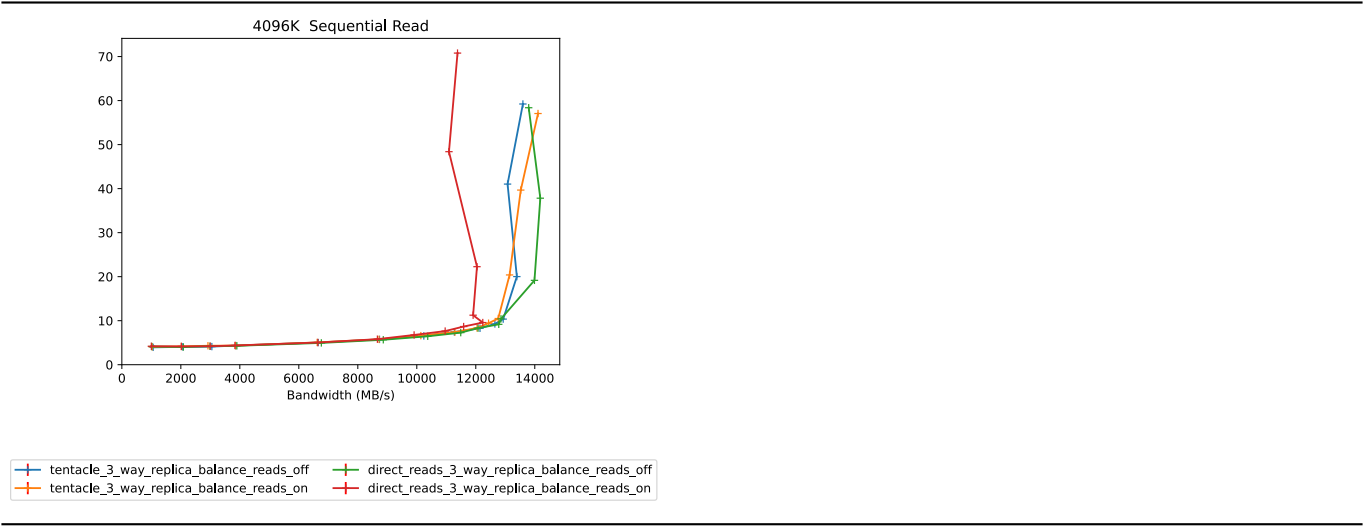
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off
tentacle_3_way_replica_balance_reads_on direct_reads_3_way_replica_balance_reads_on

tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off
tentacle_3_way_replica_balance_reads_on direct_reads_3_way_replica_balance_reads_on

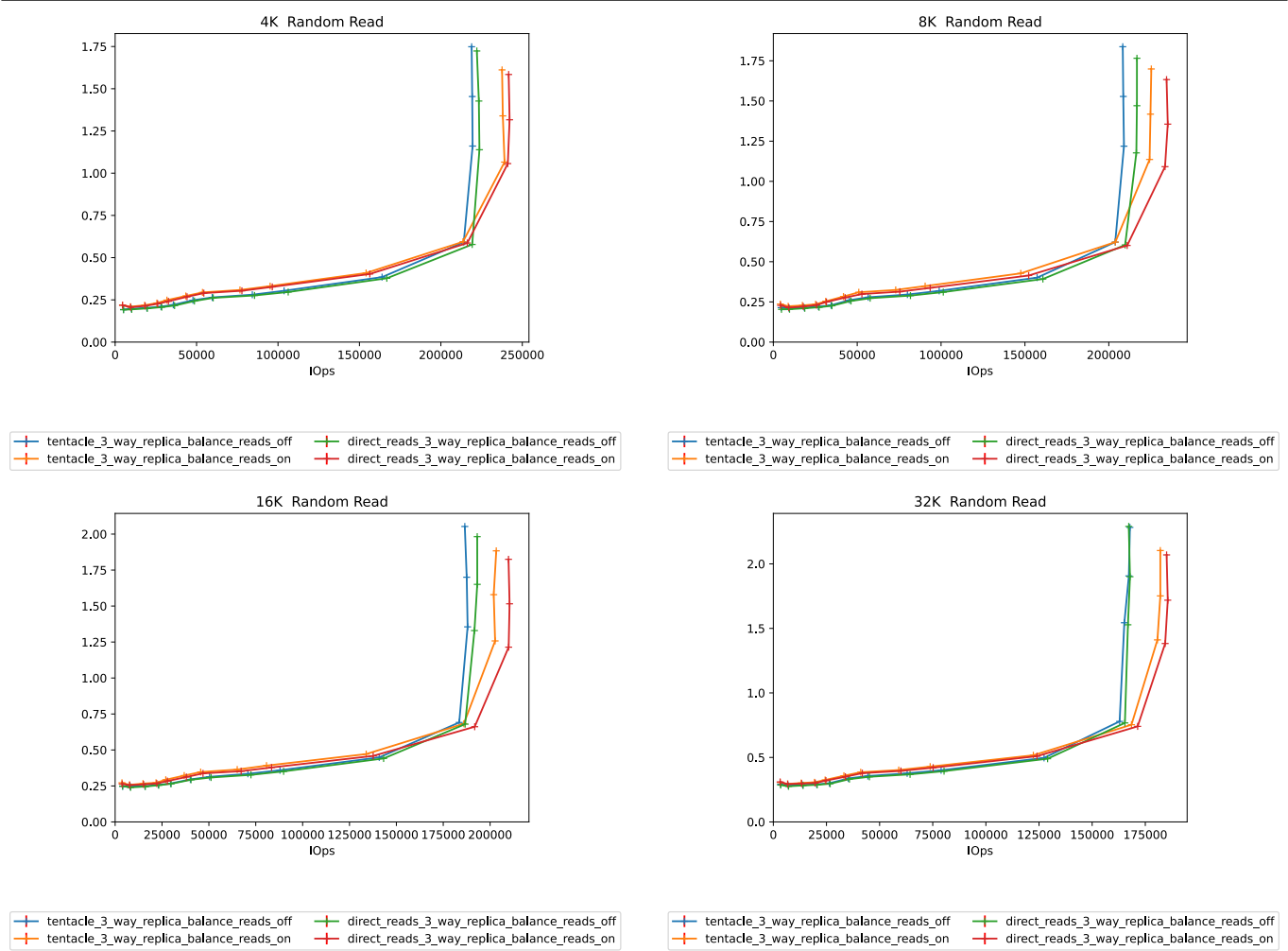


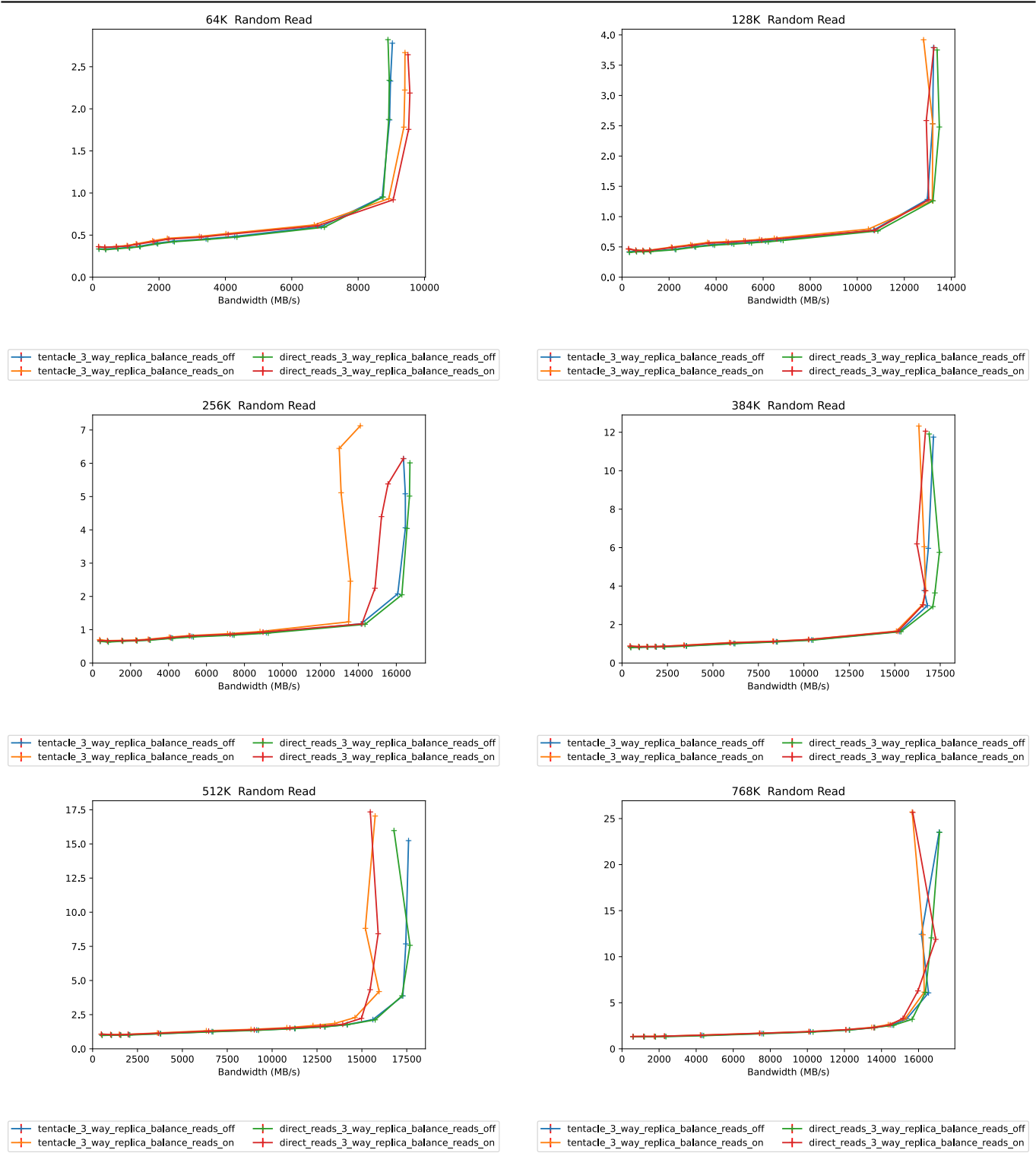
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off
tentacle_3_way_replica_balance_reads_on direct_reads_3_way_replica_balance_reads_on

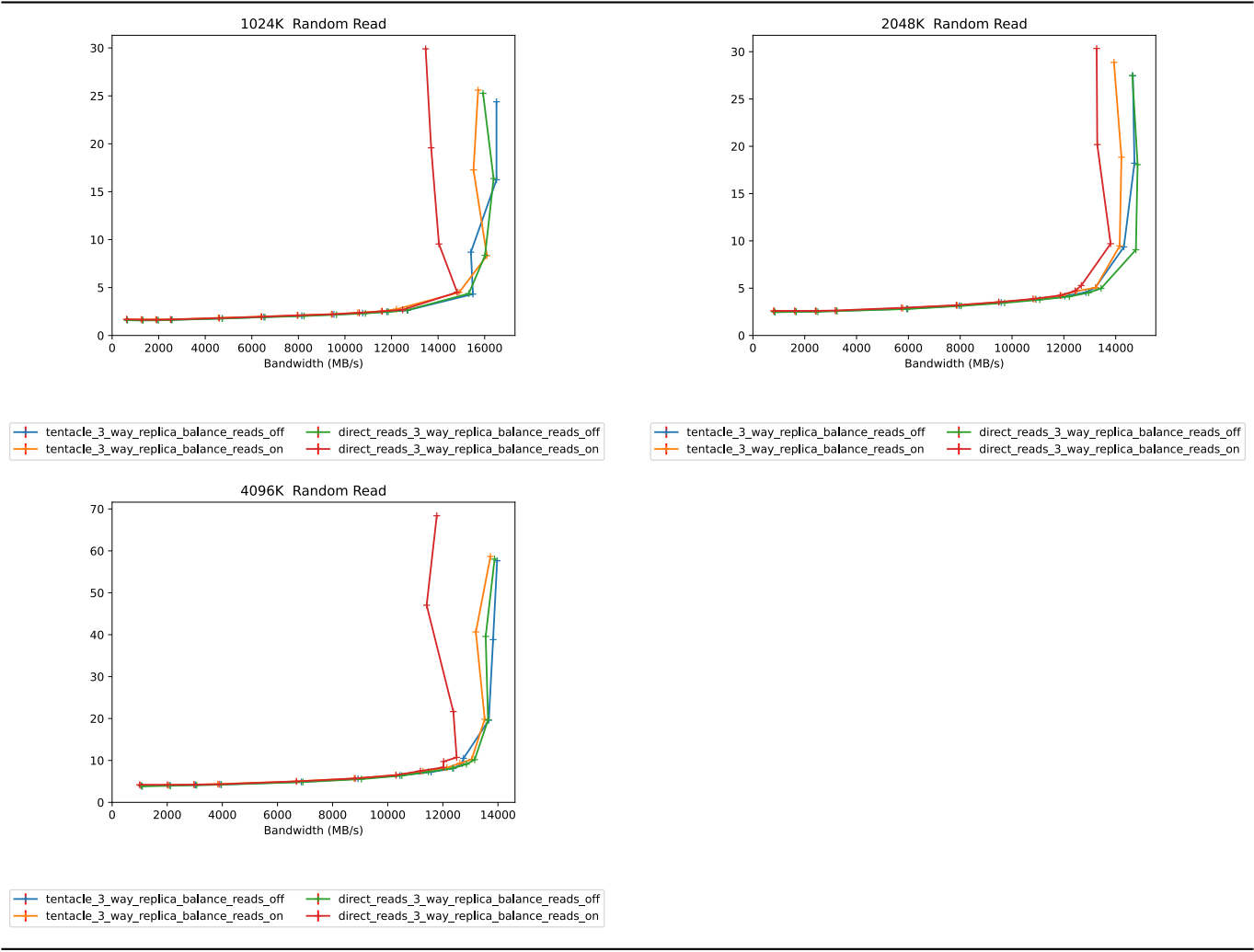
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off
tentacle_3_way_replica_balance_reads_on direct_reads_3_way_replica_balance_reads_on



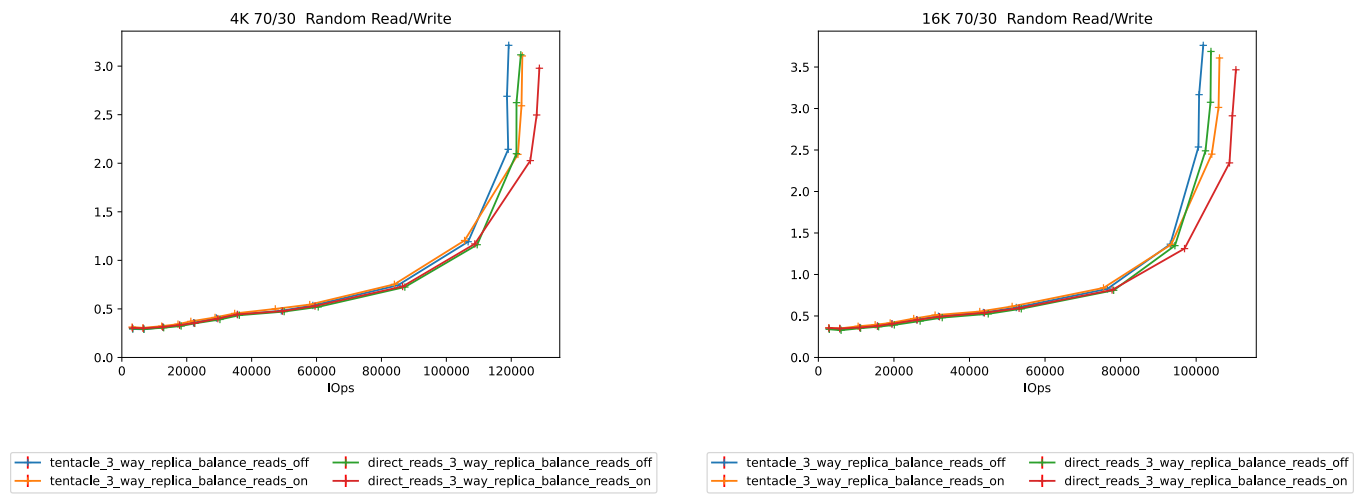
Random Read

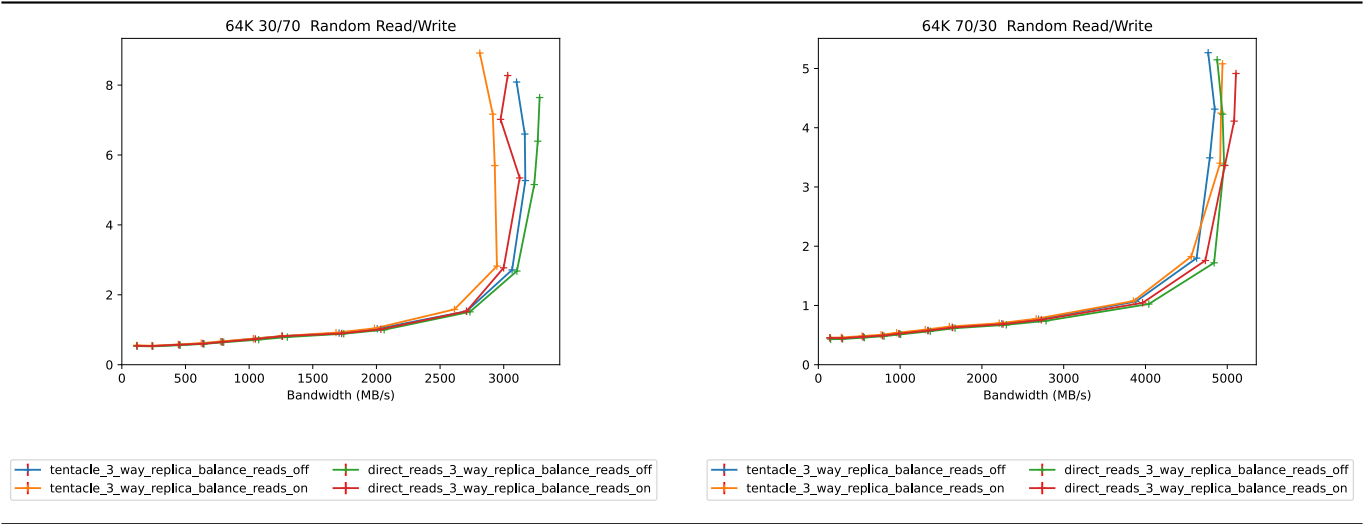






Random Read/Write





Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

1Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 1048576
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384

1Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 1048576
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384

256krandomread:

- jobname: randomread
- mode: randread
- numjobs:
- 1
- op_size: 262144
- total_iodepth:
- 1

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

256ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 262144
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

2Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

2Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

32krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

32ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

384krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

384ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1
- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

4Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4096
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4096
- total_iodepth:

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

512krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

512ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

64kmixread30:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 65536
- rwmixread: 30

```

total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1

```

```

    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  768krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 786432
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 40
      - 48
      - 64
      - 128
      - 256
      - 512
  768ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 786432
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 5
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 160
      - 256
      - 512
  8krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}
```